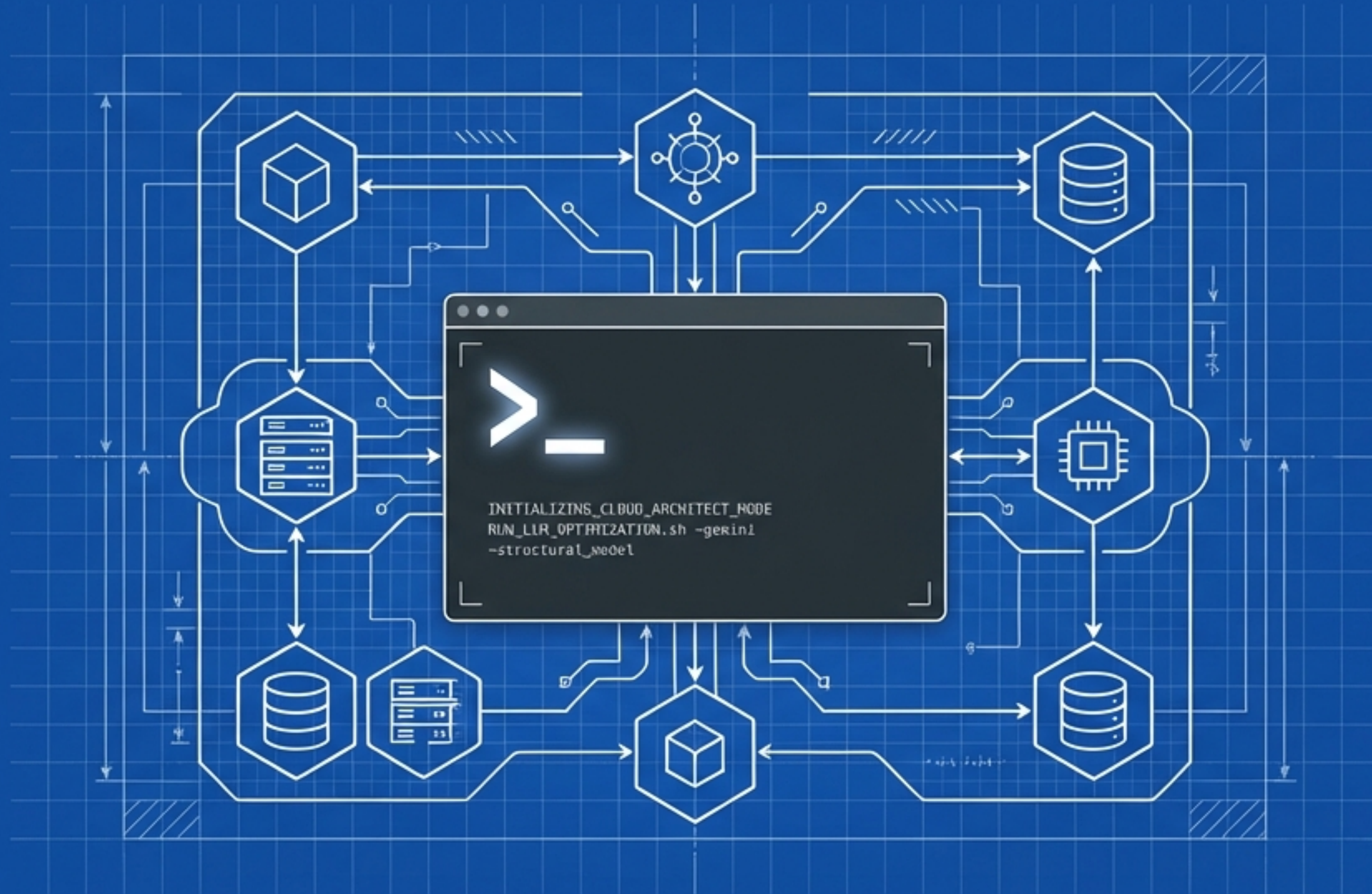


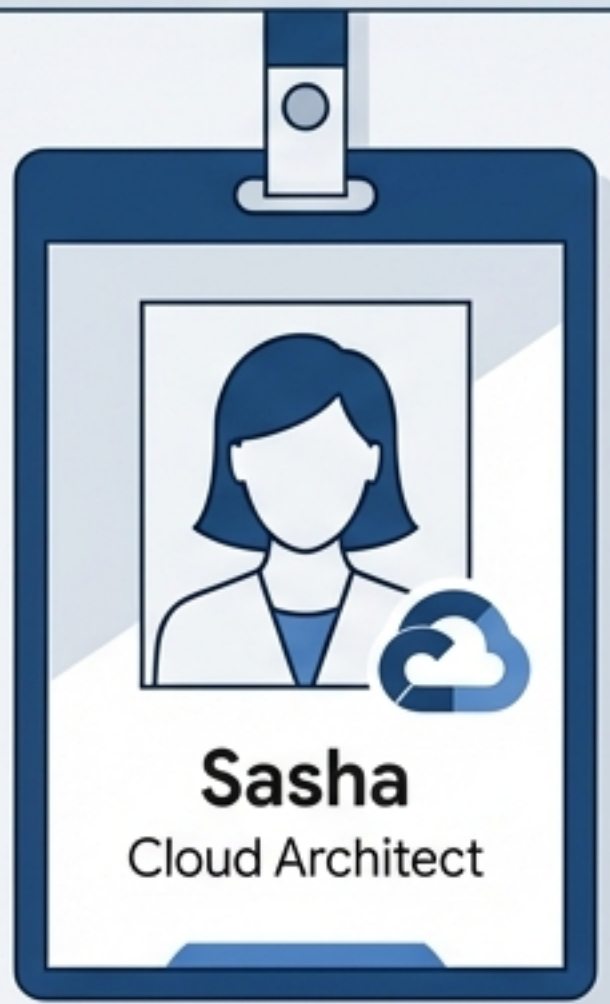


Prompt Engineering for Cloud Architects

A structural playbook for mastering Large Language Models and Gemini.

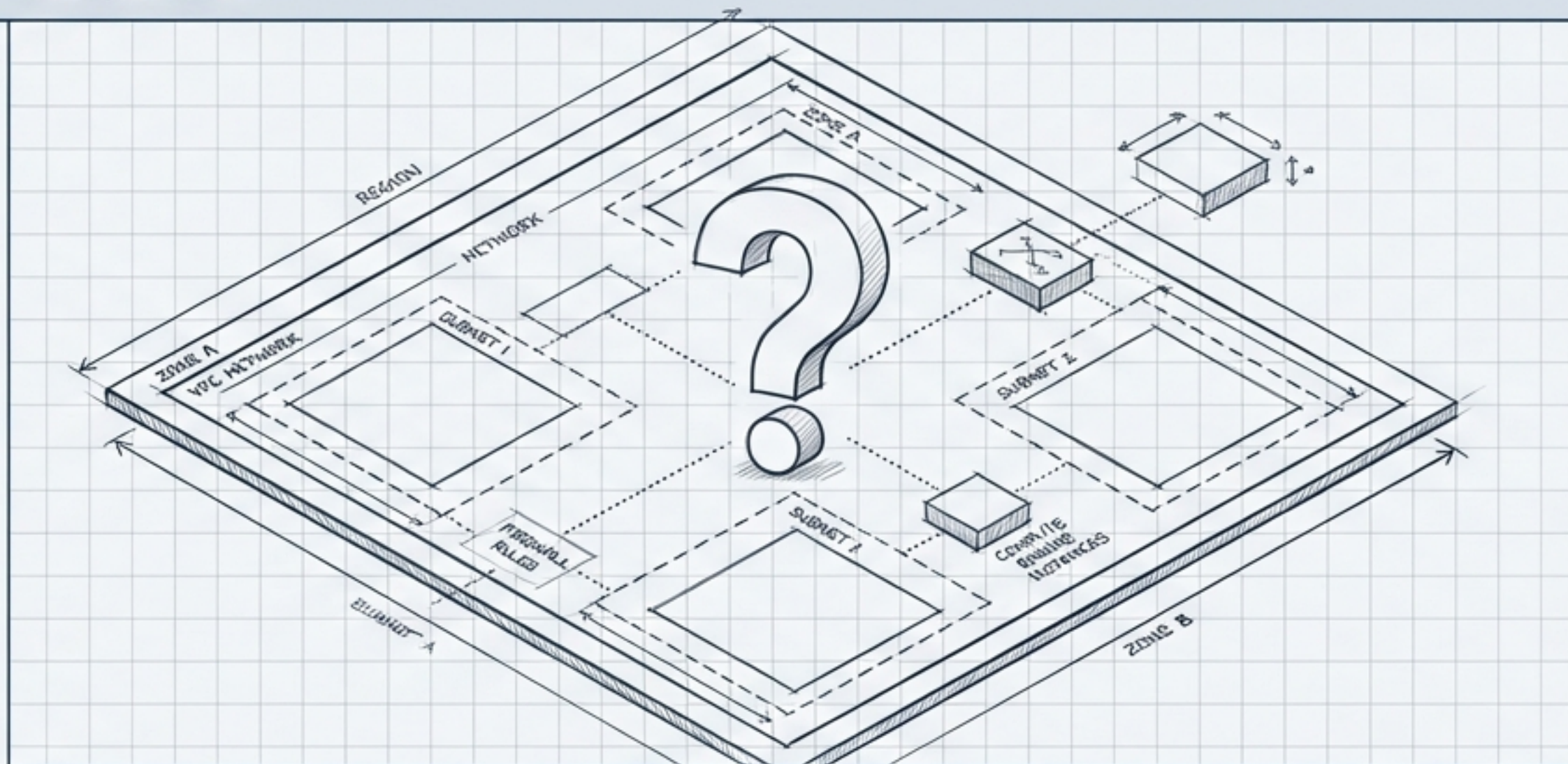
The Architecture Challenge

THE ARCHITECT



Meet **Sasha**. She is a Cloud Architect tasked with prototyping a Google Cloud VPC network architecture for Symbol Bank.

THE CANVAS



Sasha wants to save time by combining her existing cloud expertise with **Gemini**, Google's Gen AI tool, to generate a usable prototype design. To do this, she needs to master the interface between human intent and machine output.

The Technology Stack

Artificial Intelligence

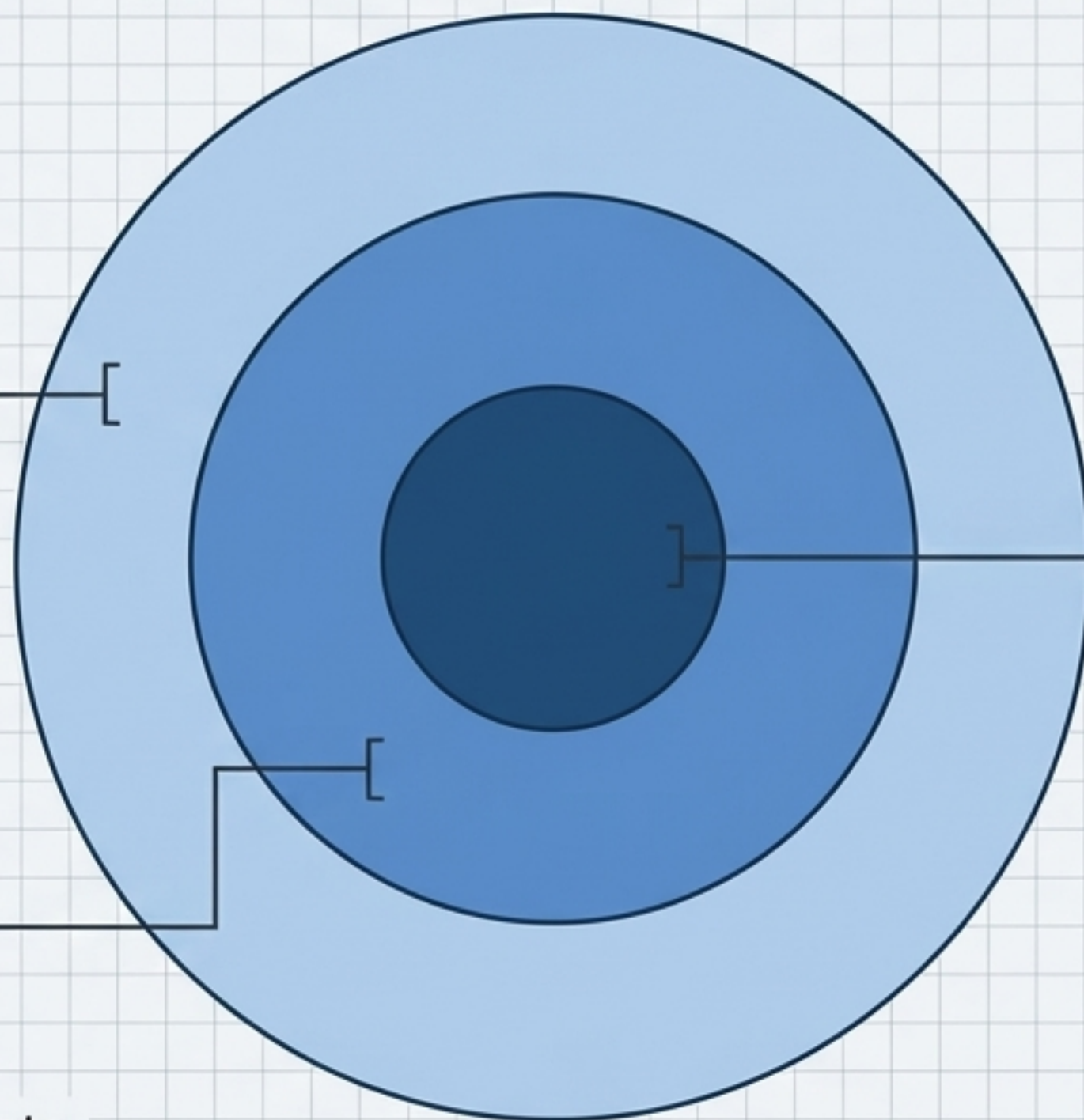
The overarching field of intelligent computing, existing since the mid-1950s.

Generative AI

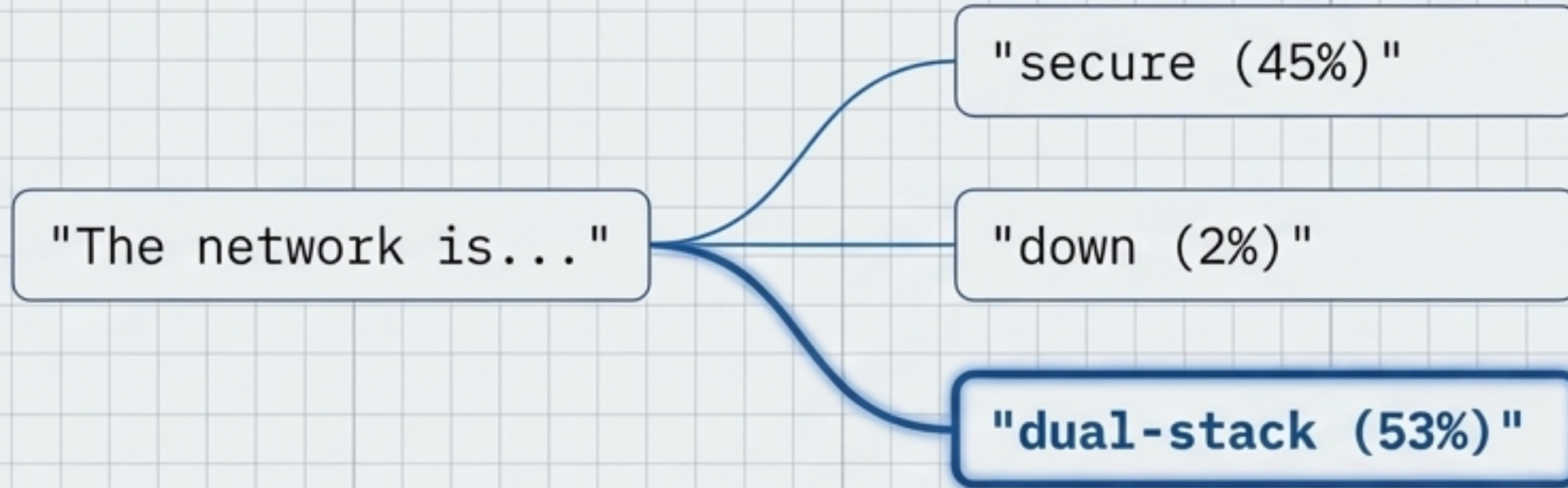
A subset capable of creating novel text, images, and data by learning patterns from training data. Exploded in popularity since 2021.

Large Language Models (LLMs)

A highly specialized subset of Gen AI focused exclusively on sophisticated human language tasks.



The Autocomplete Engine



By learning the underlying structure of language, the LLM acts as a fancy autocomplete.

It calculates the probability of the the most common correct response to your prompt.

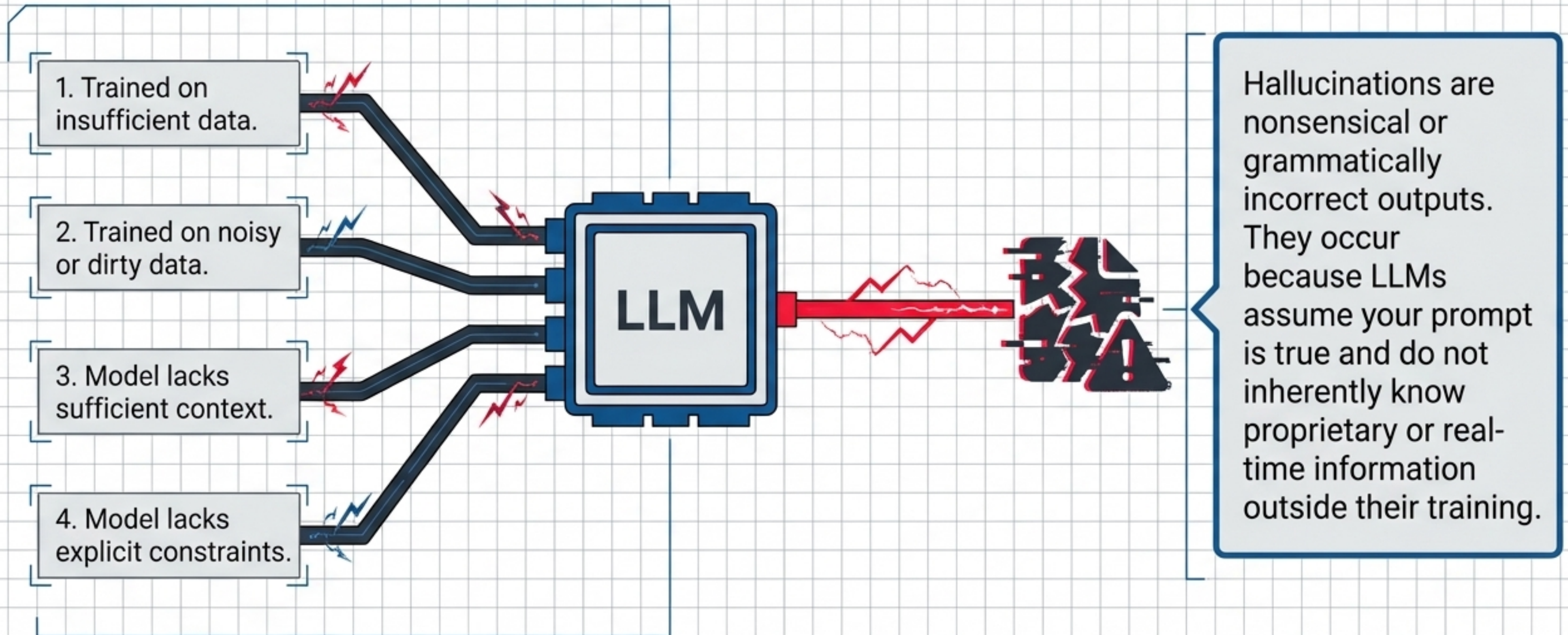


LLMs are pre-trained on massive datasets of text, images, and code. 'Large' refers to petabyte-scale data and billions or trillions of parameters—the memories and knowledge the machine learns.



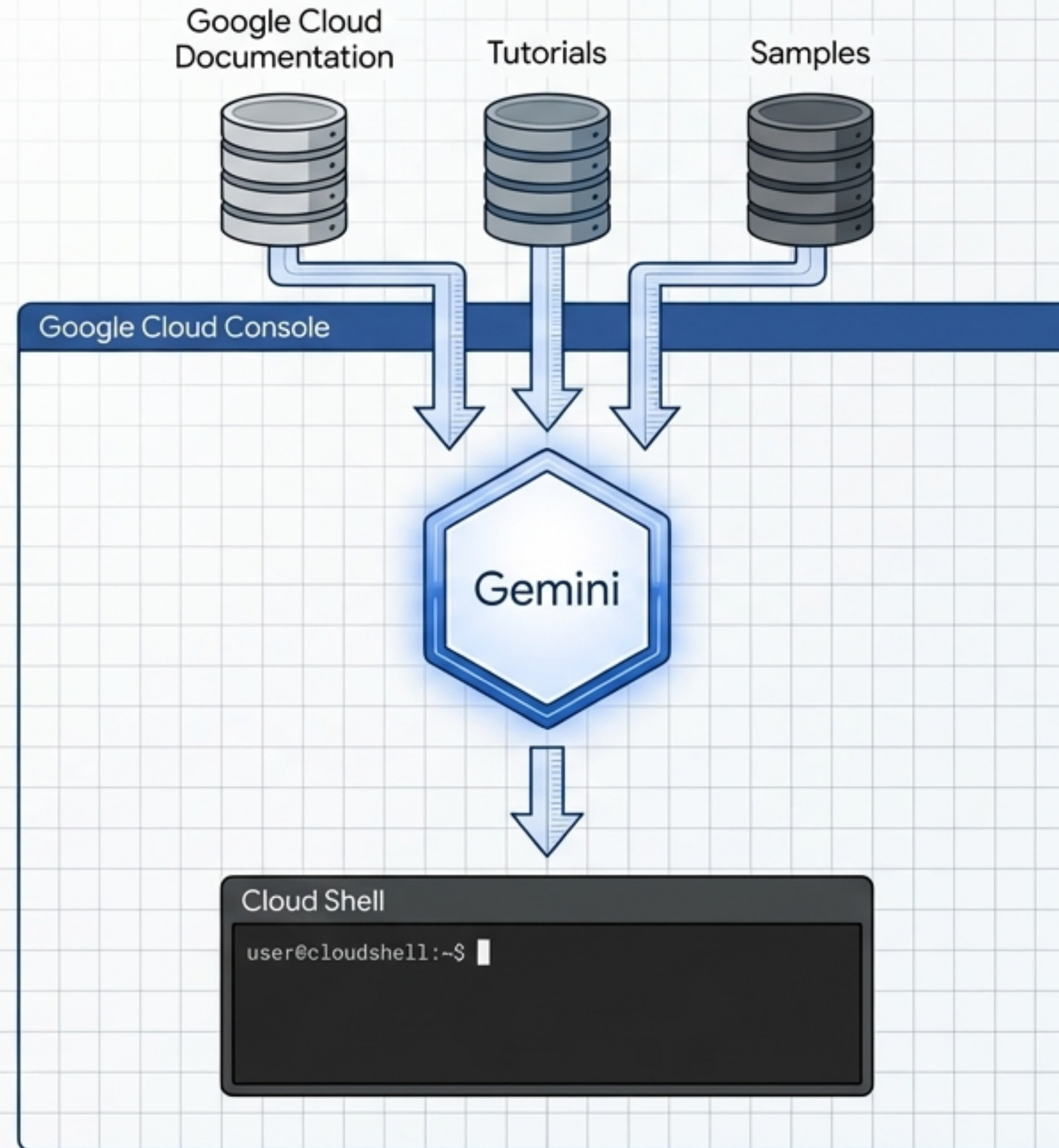
System Glitch: Understanding Hallucinations

Circuit Failure



The Always-On Collaborator

Gemini is Google's Gemini is Google's Gen AI-powered assistant, embedded directly into many Google Cloud products.

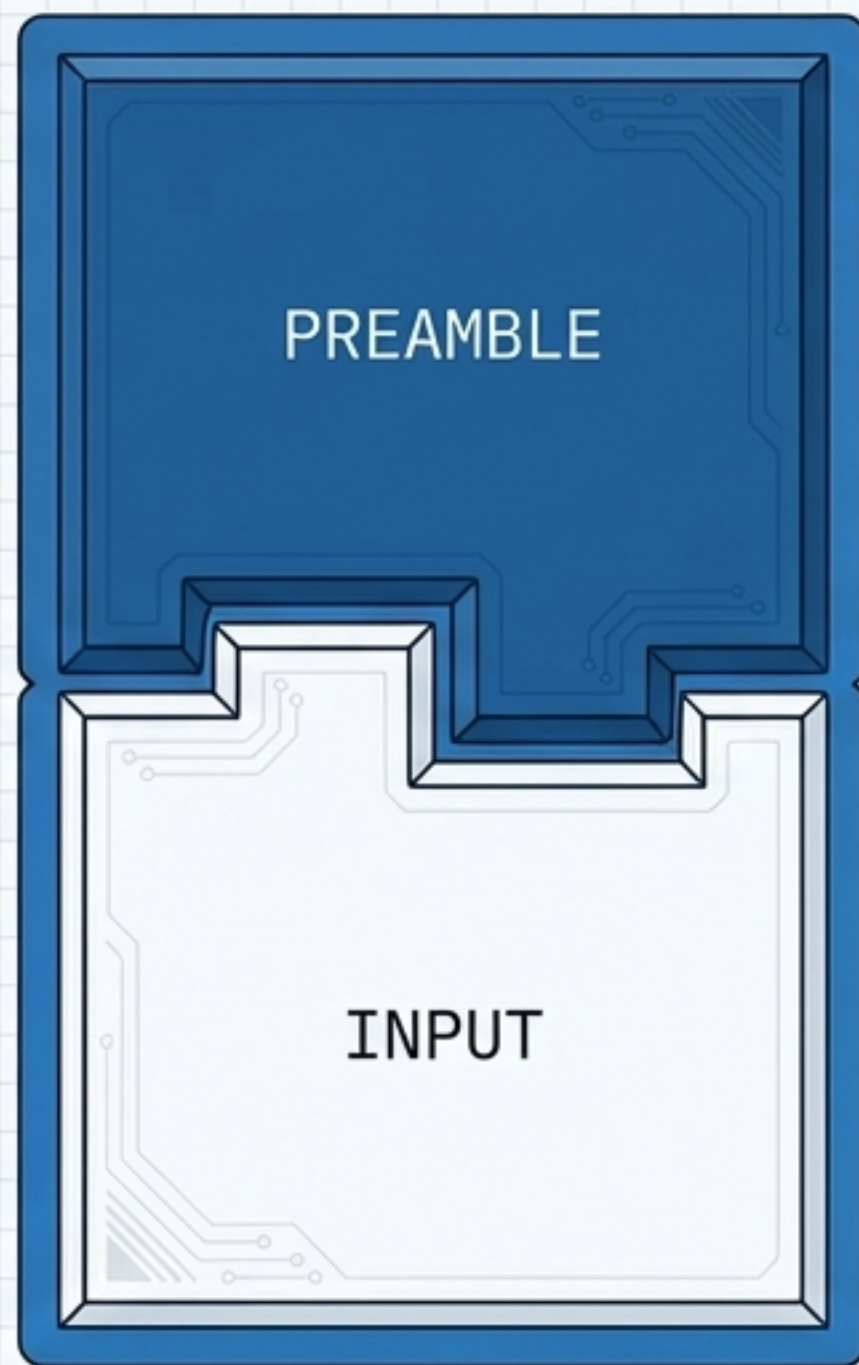


No external installations required.

Trained extensively on Google Cloud documentation.

Capable of generating detailed G-cloud commands and inserting them directly into Cloud Shell.

The Dual-Stack Prompt Anatomy



The Preamble

Introductory text that sets the stage. Contains context, task instructions, and examples.

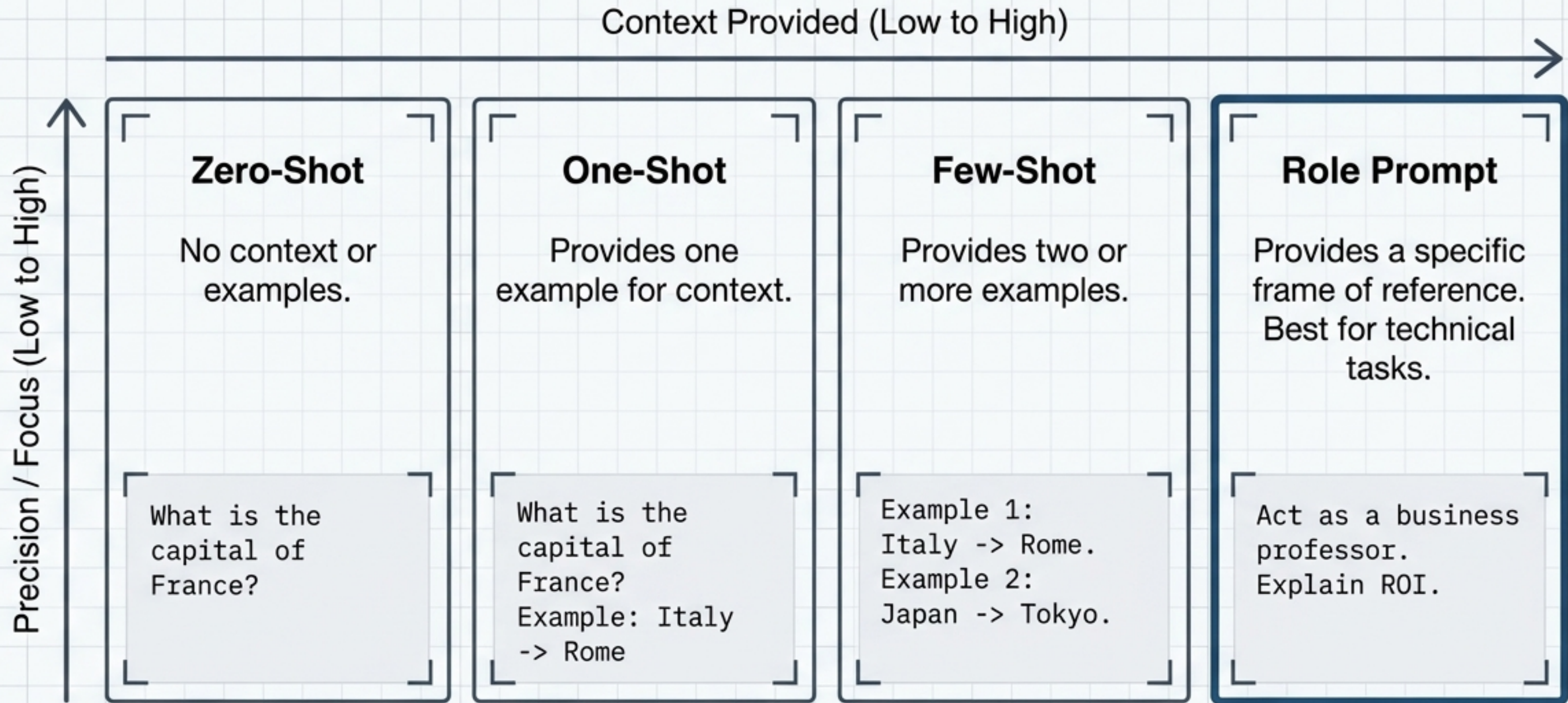
`[PREAMBLE]: You analyze user feedback. Determine if the review is positive, neutral, or negative.`

The Input

The central request. The specific item the instruction will act upon.

`[INPUT]: "I don't know what to think about this video."`

Prompting Typology Matrix



Engineering Best Practices



Write Explicit Instructions

Avoid vague language. The vaguer the prompt, the less usable the result.



Define Boundaries

Instruct the model on what to do rather than what not to do. Provide fallback outputs (e.g., 'Respond with I am still learning about that if unsure').



Adopt a Persona

Providing a meaningful role helps the model focus entirely on related context, improving accuracy.



Keep Sentences Concise

Break long, complex sentences into a series of shorter sentences and simpler, sequential tasks.

The Prompt Evolution Synthesizer

